

UPEKA DILUKSHA BSc (Hons.) Computer Engineering Undergraduate

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Built and deployed production GraphRAG and RAG systems processing 500+ page documents with semantic retrieval and graph-based entity extraction. Researching TGNN-LLM integration for real-time anomaly detection in Software Defined Vehicular Networks. Seeking AI/ML engineering roles focused on LLM orchestration, RAG pipelines, and agentic AI systems.

TECHNICAL SKILLS

Languages: Python, TypeScript / JavaScript, C/C++, SQL

AI & Machine Learning: LlamaIndex, LangChain, GraphRAG, PyTorch, Pandas, Scikit-learn, HuggingFace Transformers, Pinecone, Gemini API, OpenAI API, Vector Databases, Graph Neural Networks

Backend & DevOps: Node.js, Express.js, FastAPI, Docker, CI/CD, Neo4j, MongoDB, MySQL

Frontend: React, Next.js, TailwindCSS

EXPERIENCE

Software Engineer - Capricon Solution

2025 May - 2025 Nov

- Engineered and maintained backend REST APIs for an enterprise-scale POS system using an MVC architecture, optimizing database schemas to handle 1000+ daily transactions.
- Designed and optimized complex MySQL relational databases, applying indexing strategies to maintain high throughput data retrieval under heavy load.

Editorial Committee Lead, IEEE Student Branch, University of Ruhuna

2024 Jan - 2024 Dec

- Led the editorial committee for the IEEE student branch, producing technical publications and coordinating content strategy across events.

PROJECT EXPERIENCE

NeuroSpace - Multi-Modal Agentic AI Platform ([GitHub](#))

2026 - Present

Python, FastAPI, Neo4j, Next.js, Docker, MinIO

- Built an end-to-end GraphRAG engine capable of extracting entities from 50+ page PDFs and 1+ hour MP4 video transcripts using local embeddings.
- Engineered an Agentic GraphRAG retrieval pipeline combining Neo4j graph traversal with vector search, increasing source retrieval precision by 5.6% over vector-only baselines.
- Benchmarked LLM answer groundedness across a 50-question multi-hop evaluation set, successfully eliminating out-of-domain hallucinations by enforcing strict context-bounding and refusal protocols.
- Optimized hybrid retrieval latency, maintaining a median query response time of 4.6s while simultaneously traversing knowledge graphs and embedding spaces.

Temporal GNN-LLM Network Security - Final Year Research Project(Ongoing)

2026 - Present

Python, PyTorch, LLMs, Graph Neural Networks

- Developing an event-driven control flooding attack mitigation scheme for Software Defined Vehicular Networks (SDVN), modeling complex node interactions and routing behaviors.
- Integrating temporal graph neural networks (TGNN) with agentic LLM workflows to identify and neutralize network anomalies in real-time.
- Supervised by Dr. P.A.D.S.N. Wijesekara, focusing on applied Deep Learning for network security.

ContextIQ - Document Intelligence RAG Platform ([Live](#))

2026 Jan - 2026 March

Python, Streamlit, Gemini, Pinecone, LangChain

- Built a cloud-deployed RAG system that ingests and embeds 500+ page documents into Pinecone, managing vector embeddings for semantic retrieval.
- Engineered and iterated on prompt designs via the Gemini API, evaluating LLM outputs to optimize context retrieval and ensure system reliability across multi-turn queries.

Salary Estimator - End-to-End MLOps Pipeline ([Live](#))

2026 Jan - 2026 Feb

Python, Flask, Docker, GitHub Actions, Render

- Developed a Linear Regression salary prediction model in Python and scikit-learn, evaluating performance on held-out test data ($MAE = \$6.29K, R^2 = 0.90$).
- Productionized the model via a Flask web API with Docker containerization, ensuring system reliability.
- Engineered an automated CI/CD pipeline using GitHub Actions to run pytest validation loops prior to Render deployment.

EDUCATION

BSc (Hons.) in Engineering - Computer Engineering

2021 - Present

University of Ruhuna - Faculty of Engineering, Graduating August 2026 - available immediately after.

G.C.E. Advanced Level (A/L) - Physical Science Stream

2019

Mahinda College Galle, Result: AAB - District Rank 126

LICENSES & CERTIFICATIONS

[Generative AI with Large Language Models](#): Issued by DeepLearning.AI

[Supervised Machine Learning](#): Issued by DeepLearning.AI

[Machine Learning in Production](#): Issued by DeepLearning.AI

[Generative AI: Prompt Engineering Basics](#): Issued by IBM

[Introduction to DevOps](#): Issued by IBM

[Introduction to Artificial Intelligence \(AI\)](#): Issued by IBM